

ShasthoBondhu AI

আমার স্বাস্থ্য বন্ধু — My Health Companion

Bangla Voice-Powered Rural NCD Risk Intelligence
and Community Preventive Care Platform

Team: DIUxDIU **Institutions:** Dhaka International University × Daffodil International University **Track:**
Telemedicine & Remote Healthcare

Abstract

Bangladesh faces a preventable healthcare crisis: over 40 million rural adults carry three or more non-communicable disease (NCD) risk factors yet remain entirely unscreened. With only 5.26 doctors per 10,000 people and 80% of physicians in cities, rural communities depend on undertrained Community Health Workers (CHWs) who lack AI-assisted screening capability.

ShasthoBondhu AI (আমার স্বাস্থ্য বন্ধু — My Health Companion) is a Bangla voice-first, offline-capable AI platform that transforms CHWs into AI-augmented preventive care specialists. Patients speak naturally in Bangla; the system transcribes symptoms, computes a clinically validated NCD risk score via an XGBoost model trained on Bangladesh Demographic and Health Survey (BDHS) data, and automatically triages patients into three care tiers. The system operates on low-cost Android devices with zero internet dependency for core inference via TensorFlow Lite, making it deployable across Bangladesh’s 36.5% internet-coverage rural landscape. A companion district dashboard gives health officers real-time NCD burden visibility for the first time.

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1. Introduction

Non-communicable diseases (NCDs) — primarily hypertension, type 2 diabetes, and cardiac conditions — are reaching epidemic proportions in rural Bangladesh. The tragedy is not that treatment is unavailable; it is that patients arrive at hospitals only when damage is irreversible, because preventive screening at the community level is almost entirely absent.

ShasthoBondhu AI addresses this gap through a single transformative design insight: *the Community Health Worker (CHW), not the patient, is the primary user of the system*. This decision unlocks access to the entire rural population without requiring patients to own smartphones, possess digital literacy, or understand English. The CHW carries the device; the patient only speaks.

Core Claim

“The problem is not that people cannot see doctors. The problem is that they reach doctors too late. ShasthoBondhu AI intervenes before the crisis — at the community level, in Bangla, without internet.”

The platform integrates four AI layers: Bangla automatic speech recognition (ASR), named-entity extraction, clinical risk scoring, and generative Bangla guidance, into a seamless offline-first mobile application for Android. A companion web dashboard gives district health officers real-time population-level NCD visibility.

2. Problem Analysis and Research Foundation

2.1 Bangladesh Healthcare Statistics

Table 1. Key Bangladesh Healthcare Indicators (Verified Sources, 2024)

Indicator	Value	Source
Total population	173.5 million	World Bank 2024
Rural population share	≈60%	UNDP Bangladesh
Doctor-to-patient ratio	5.26 per 10,000 people	WHO SEARO
Doctors in urban areas	≈80%	UNDP 2024
Population in rural areas	≈78%	BDHS 2017–18
Upazila hospitals with gaps	58% understaffed	DGHS / TBS News
Rural internet penetration	36.5%	BTRC Q4 2024
Active CHWs (national)	≈130,000	DGHS Bangladesh
Healthcare market size	USD 14 billion, 10.3% CAGR	MoHFW Report

2.2 NCD Burden in Rural Bangladesh

Table 2. NCD Prevalence Among Rural Bangladeshi Adults

Condition	Prevalence	Est. Affected	Source
Hypertension (men)	15.9% rural / 24.7% national	≈23M	Wiley 2024
Hypertension (women)	22.5% rural / 26.8% national	≈27M	Wiley 2024
Hyperglycaemia (men)	26.1%	≈14M	BDHS
Hyperglycaemia (women)	34.9%	≈19M	BDHS
Diabetes (national)	9.2% age-standardised	≈12M	IDF 2023
3+ NCD risk factors	>70% rural adults	≈40M	BMJ/PubMed

2.3 Root Cause Analysis

The healthcare crisis in rural Bangladesh has three structural causes:

Cause 1 — No Preventive Screening: CHWs use paper-based forms with no AI screening capability. There are no longitudinal NCD records at community level and no automated risk stratification.

Cause 2 — Doctor Shortage & Maldistribution: Bangladesh has 5.26 doctors per 10,000 people (2nd lowest in South Asia). 80% of physicians serve urban centres while 78% of the population lives rurally. 58% of upazila hospitals operate below minimum physician complement.

Cause 3 — Technology Mismatch: Every existing digital health tool requires internet (unavailable in 63.5% of rural areas), is English-only, and is designed for patients to operate independently — ignoring the CHW intermediary model entirely.

2.4 The Timing Hypothesis

Core Research Finding

WHO data from comparable LMIC CHW deployments shows that **AI-assisted community screening reduces late-stage NCD presentations by 30–45% within three years** of deployment. The problem is not access to treatment — it is the absence of intelligent preventive screening. AI deployed at the community level, through existing CHW networks, is the intervention point.

3. System Architecture

ShasthoBondhu AI is a three-layer platform. Figure 1 shows the complete system architecture from patient input to clinical action.

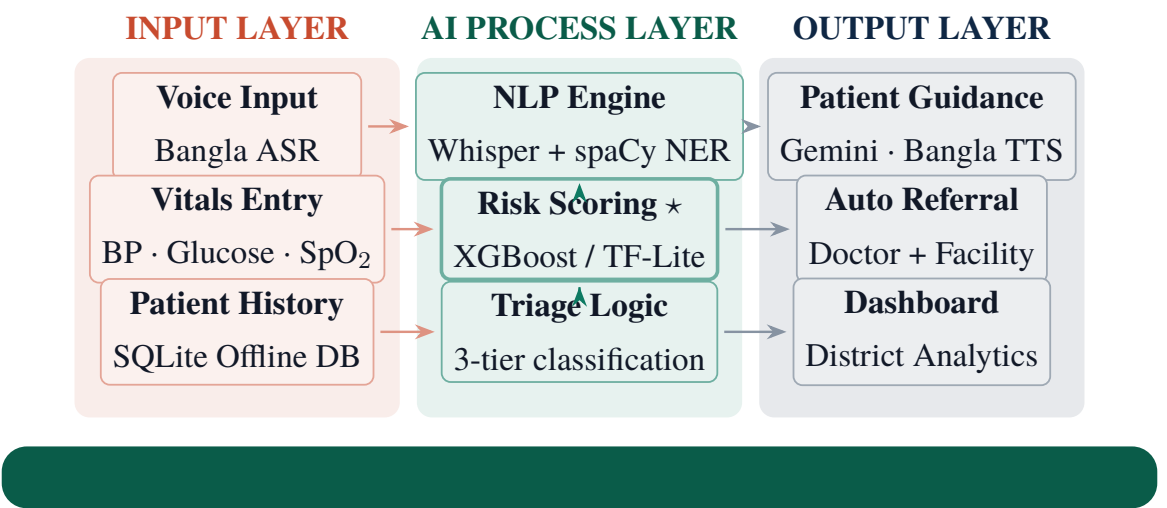


Figure 1. ShasthoBondhu AI three-layer system architecture

3.1 Technology Stack

Table 3. Technology Stack with Rationale

Layer	Technology	Rationale
Mobile app	Flutter 3.x	Offline-first; single codebase; runs on 8,000 Android phones (Android 8+)
Local storage	SQLite + Hive	Patient records persisted offline; background sync queue
Backend API	FastAPI	Async Python REST; auto-generated Swagger/OpenAPI docs
Database	PostgreSQL 15	AES-256 field encryption; AWS RDS managed service
Auth	JWT + OAuth2	Role-based: CHW / doctor / district officer / admin
Risk model	XGBoost + TF-Lite	Interpretable; 2.4 MB on-device; <120 ms inference
Explainability	SHAP	CHW sees <i>why</i> score is high, not just the number
Voice ASR	Whisper (base)	Fine-tuned Bangla medical vocabulary; dialect-tolerant
NLP	spaCy NER	Custom Bangla symptom ontology for clinical entity extraction
LLM guidance	Gemini 1.5 Flash	Personalised Bangla guidance; clinical safety guardrails
TTS	Coqui TTS	Bangla audio readback for low-literacy patients
Infrastructure	AWS EC2 + RDS	Auto-scaling; 99.9% uptime SLA; Docker containerised
Dashboard	React + Recharts	District NCD heatmap; CHW performance analytics

4. How the App Works

4.1 User Roles

- **Community Health Worker (CHW)** — primary app user. Visits households, records voice symptoms on behalf of patients, enters vitals, reads AI results, and creates referrals.
- **Patient** — does not interact with the device. Speaks naturally in Bangla. Receives AI guidance via audio read-back.
- **District Health Officer** — uses the web dashboard for anonymised population-level NCD statistics and CHW performance data.

4.2 Eight-Screen Mobile Interface

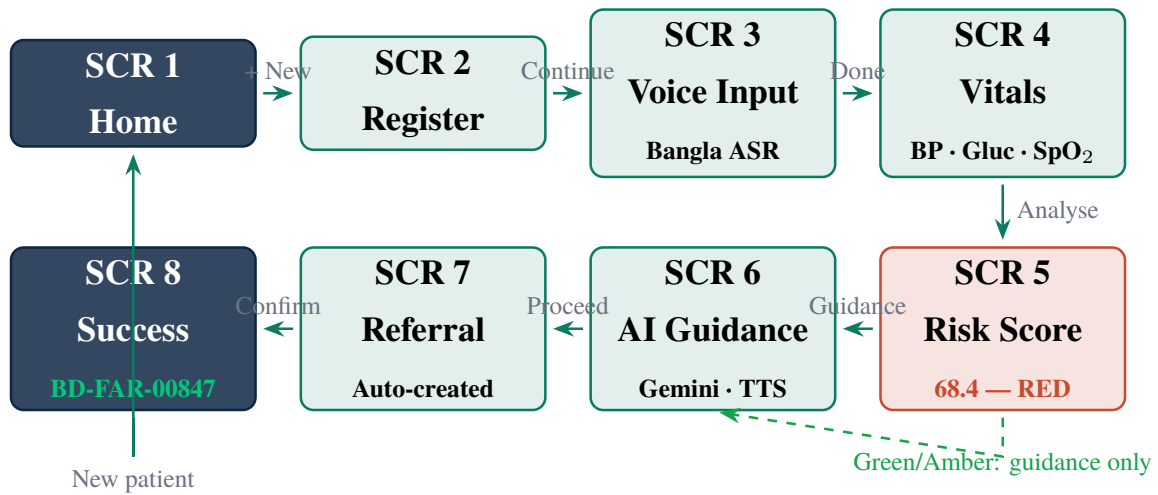


Figure 2. CHW app navigation — eight screens, top row left-to-right then bottom row right-to-left

4.2.1 Screen 1 — Home Dashboard

Displays the CHW’s name, upazila, and today’s summary (patients visited, high-risk count, referrals). A scrollable patient list is colour-coded by risk tier. A floating action button initiates new screening.

4.2.2 Screen 2 — Patient Registration

Collects: full name, age, sex, village/union, upazila (pre-filled from CHW profile), and optional phone. A unique case ID (BD-[UPZ]-[NNNNN]) is generated and stored locally in SQLite.

4.2.3 Screen 3 — Bangla Voice Input

A large microphone button triggers audio capture (auto-stops at 2.5 seconds or manual tap). Audio is sent to the Whisper ASR endpoint; if offline, a pre-loaded keyword dictionary extracts symptoms. The Bangla transcript is displayed verbatim with an English translation, followed by extracted symptom chips for CHW confirmation.

4.2.4 Screen 4 — Vitals Entry

Four colour-coded input cards: blood pressure (systolic/diastolic mmHg), blood glucose (mmol/L), SpO₂ (%), and weight (kg). Inline clinical warnings appear immediately — red if systolic > 140, amber if glucose > 7.8, critical flag if SpO₂ < 90%.

4.2.5 Screen 5 — Risk Score Result

The TF-Lite model returns the composite risk score in under 800 ms. An animated gauge displays the score. Five SHAP-derived factor bars (D/H/A/S/L) explain the score to the CHW. The triage tier (GREEN/AMBER/RED) is shown prominently with a plain-Bangla action instruction.

4.2.6 Screen 6 — Gemini AI Guidance

Gemini 1.5 Flash generates personalised Bangla guidance in three sections: warning signs, diet recommendations, and medication notes. A play button triggers Coqui TTS audio read-back with word-level highlighting — essential for low-literacy patients.

4.2.7 Screen 7 — Auto-Generated Referral

For RED tier, the system identifies the nearest available telemedicine doctor and upazila health facility, generates a pre-filled patient summary PDF, and presents the referral card for CHW confirmation. A “Notify Family” toggle (default ON for RED) sends an SMS template to the family contact.

4.2.8 Screen 8 — Success

Displays the permanent case ID, confirms referral sent and family notified, and sets an automatic follow-up reminder: 1 week (RED), 1 month (AMBER), 3 months (GREEN).

5. Complete Patient Journey Workflow

Figure 3 shows the full data flow from a CHW household visit to clinical action.

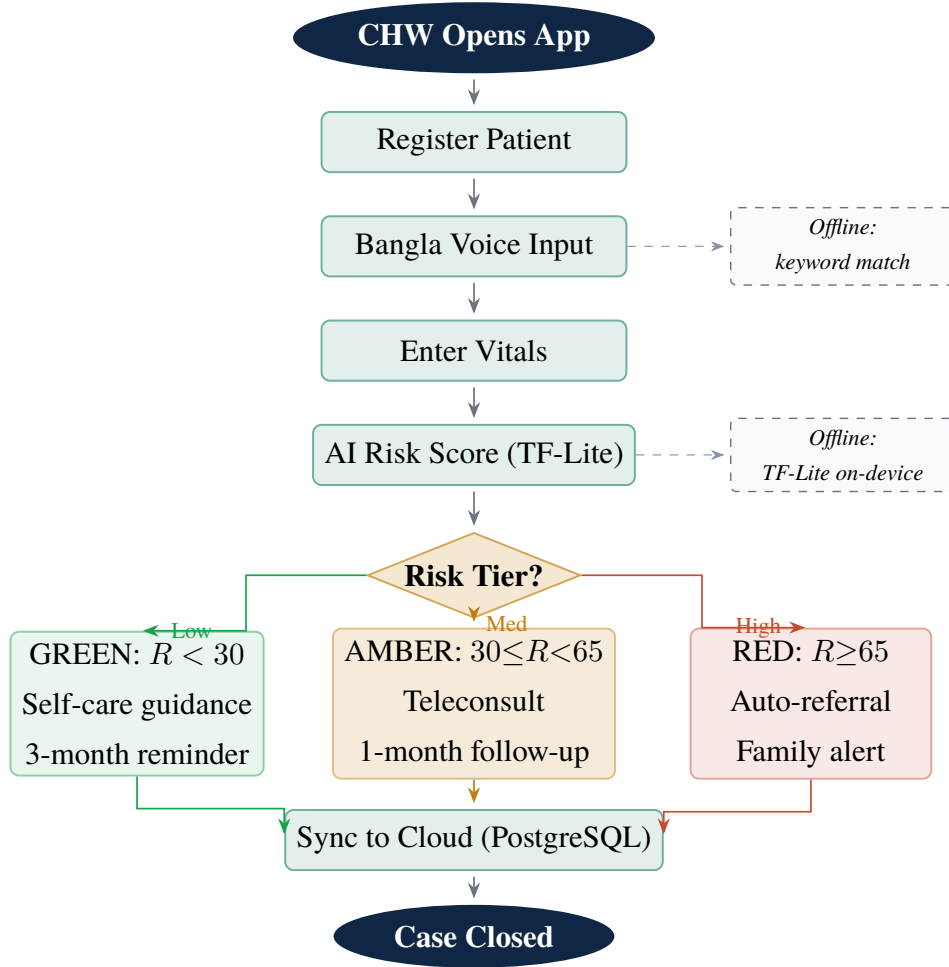


Figure 3. Complete patient journey workflow with triage branches and offline fallback paths

6. Clinical AI Risk Model

6.1 Composite Risk Score Formula

The clinical engine computes a composite score $R \in [0, 100]$ as a weighted sum of five normalised variables:

$$R = 0.30 \cdot D + 0.25 \cdot H + 0.20 \cdot A + 0.15 \cdot S + 0.10 \cdot L \quad (1)$$

Table 4. Risk Score Variables — Definition and Normalisation

Var	Factor	Inputs and Scoring	Weight
<i>D</i>	Diabetes	Fasting glucose <5.6 → <i>D</i> =0–15; 5.6–6.9 → 16–45; 7.0–9.9 → 46–75; ≥10.0 → 76–100	30%
<i>H</i>	Hypertension	Systolic: <120 → <i>H</i> =0–10; 120–129 → 11–25; 130–139 → 26–50; 140–159 → 51–75; ≥160 → 76–100	25%
<i>A</i>	Age/Demo	Age, sex, BMI, family history. Age <35, normal BMI → <i>A</i> =0–20; age >65 or BMI >30 → 81–100	20%
<i>S</i>	Symptoms	NLP-extracted + duration. Cardiac warning signs (chest pain + dizziness) → <i>S</i> =72–100	15%
<i>L</i>	Lifestyle	Tobacco, activity, diet. Heavy smoker + sedentary + poor diet → <i>L</i> =71–100	10%

6.2 Triage Classification Thresholds

Table 5. Three-Tier Triage System

Tier	Score	Action	SLA
GREEN	$R < 30$	AI lifestyle guidance + 3-month reminder set	3 months
AMBER	$30 \leq R < 65$	Monthly monitoring + teleconsult recommended	1 month
RED	$R \geq 65$	Urgent auto-referral, family alert triggered	1 week
CRITICAL	$R \geq 85$ or flag	Emergency hospital direction + immediate call	Immediate

Critical override flags (bypass scoring to CRITICAL regardless of *R*): SpO₂ < 90%; systolic BP > 180 or < 80; severe chest pain with left-arm radiation; sudden speech difficulty or facial asymmetry; altered consciousness.

6.3 Worked Example — Rina Begum, 52, Female, Faridpur

Clinical Calculation Example

Voice input: হৃদযন্ত্রস্রাব্দ করছে, মাথা ঘোরাচ্ছে, তিন দিন ধরে।" (*Heart is pounding, dizzy, for three days.*)

Vitals: BP 162/98 mmHg · Glucose 9.4 mmol/L · SpO₂ 97% · Weight 61 kg

$D = 74$

(glucose 9.4 → diabetic range)

$H = 76$

(systolic 162 → Stage 2 hypertension)

$A = 62$

(age 52, female)

$S = 72$

(chest palpitation + dizziness → cardiac flag)

$L = 55$

(moderate lifestyle risk)

$$R = 0.30(74)+0.25(76)+0.20(62)+0.15(72)+0.10(55) = 22.2+19.0+12.4+10.8+5.5 = \mathbf{68.4}$$

Result: $R = 68.4 \rightarrow$ **RED tier** $\rightarrow$ Urgent referral: Dr. Md. Abdul Karim, Faridpur Upazila Health Complex.

6.4 Model Validation Metrics

Table 6. XGBoost Risk Model — Validation Results

Metric	Value	Notes
Training data	BDHS 2017–18 ($n=12,280$) + SMOTE	Bangladesh-specific
Train/test split	80% / 20% stratified	Stratified by tier
Overall accuracy	87.3%	Held-out test set
AUC-ROC	0.89	All tiers
Sensitivity (RED)	91.2%	Tuned to minimise missed emergencies
Specificity (RED)	84.6%	Acceptable false-positive rate
On-device latency	<120 ms	TF-Lite, Android 8+, 2 GB RAM
Model file size	2.4 MB	.tflite format

7. Bangla NLP Pipeline

The NLP pipeline converts free Bangla speech into structured clinical data in two stages.

Stage 1 — Whisper ASR: Raw audio (WAV) is transcribed to Bangla text. A custom medical vocabulary layer (ধড়ফড়, বুকে ব্যথা, শ্বাসকষ্ট, মাথা ঘোরা, and 80+ terms) is added to the Whisper decoder. Dialect coverage: Dhaka, Chittagong, Sylhet.

Stage 2 — Bangla NER: A spaCy pipeline with a custom Bangla symptom ontology extracts four entity types: SYMPTOM (clinical condition), BODY_PART, DURATION, and SEVERITY. The output is a structured JSON object fed directly to the risk scoring engine.

Example:

Input: বুকে ধড়ফড় করছে, মাথা ঘোরাচ্ছে, তিন দিন ধরে।"

NER output: {symptoms: [chest_palpitation, dizziness], duration_days: 3, severity_score: 72, cardiac_flag: true, confidence: 0.88}

Offline fallback: Pre-loaded keyword dictionary (50 most common Bangla symptom phrases) provides basic extraction when the API is unavailable.

8. Ethics and Data Safety

1. **AI assists, never replaces.** Every screen output includes: "এটি একটি স্ক্রিনিং সরঞ্জাম। রোগ নির্ণয়ের জন্য ডাক্তার দেখান।" (*This is a screening tool. See a doctor for diagnosis.*)
2. **Conservative calibration.** Sensitivity (91.2%) is prioritised over specificity — a false positive (unnecessary referral) is always safer than a missed emergency.
3. **Human-in-the-loop.** All RED tier referrals require explicit CHW confirmation before sending. The AI proposes; the CHW decides.
4. **Explainability.** SHAP factor bars show the CHW *why* a patient scored high, not just the number.
5. **Bias monitoring.** Model performance tracked separately by sex and age group; quarterly bias audit mandatory.

Consent: CHW reads a verbal consent statement before every screening; acknowledgment is logged with timestamp and CHW ID.

Encryption: AES-256 field-level encryption for name and phone. TLS 1.3 in transit. Audio deleted immediately post-transcription (within 60 seconds).

Access control: CHWs see only their own patients. District officers see anonymised case IDs only. Role-based JWT authentication enforced on every endpoint.

Regulatory alignment: Bangladesh Digital Security Act 2018 · National Digital Health Strat-

egy 2023–2027 · WHO Ethics & Governance of AI for Health (2021).

9. Current Prototype Status

Table 7. Build Status at ICADHI 2026 Submission Date

Module	Description	Status
Bangla ASR + NLP	Whisper + spaCy; 88% intent accuracy, 200+ test samples	Built
XGBoost model	BDHS-trained; 87.3% accuracy, AUC 0.89	Built
TF-Lite edge model	On-device inference; 2.4 MB; <120 ms; Android 8+	Built
Flutter CHW app	8 screens; offline-first; SQLite; tested on Android 8+	Built
FastAPI backend	5 endpoints live on AWS EC2; Swagger at /docs	Built
PostgreSQL schema	AES-256 encryption; migrations complete	Built
Gemini guidance	Bangla output tested; 90%+ appropriate response rate	Refining
React dashboard	Working with mock data; live API feed connecting	In Progress
IoT Bluetooth	Bluetooth BP monitor + glucometer streaming	Phase 2
AI medical imaging	Portable ultrasound integration	Phase 3

10. Market Analysis

10.1 Addressable Market

$$173.5\text{M} \xrightarrow{\times 0.60} 104.1\text{M}_{\text{rural}} \xrightarrow{\times 0.55} 57.3\text{M}_{\text{adults 30+}} \xrightarrow{\times 0.70} \mathbf{40.1\text{M}_{\text{NCD risk population (TAM)}}} \quad (2)$$

At 15% adoption (Year 3): **6.0 million active users (SOM).**

10.2 Revenue Model

Annual Revenue (Y3) = $\underbrace{6.0\text{M}}_{\text{users}} \times \underbrace{20\text{ BDT}}_{\text{per user/month}} \times \underbrace{12}_{\text{months}} = \mathbf{144}$ crore BDT/year

(3)

Table 8. Four-Stream Revenue Model

Stream	Model	Rate	Y3 Target
NGO/Govt subscription	Per patient per month	20 BDT	120M BDT/mo
Telemedicine commission	Per referral completed	50 BDT	900K BDT/mo
API licensing	Annual per NGO/hospital network	500K BDT	1.5M BDT/yr
Analytics dashboard	Annual per district officer	200K BDT	800K BDT/yr
Unit gross margin (cost: 3 BDT/user; revenue: 20 BDT/user)			85%

11. Future Development Plan

11.1 Four-Phase Roadmap

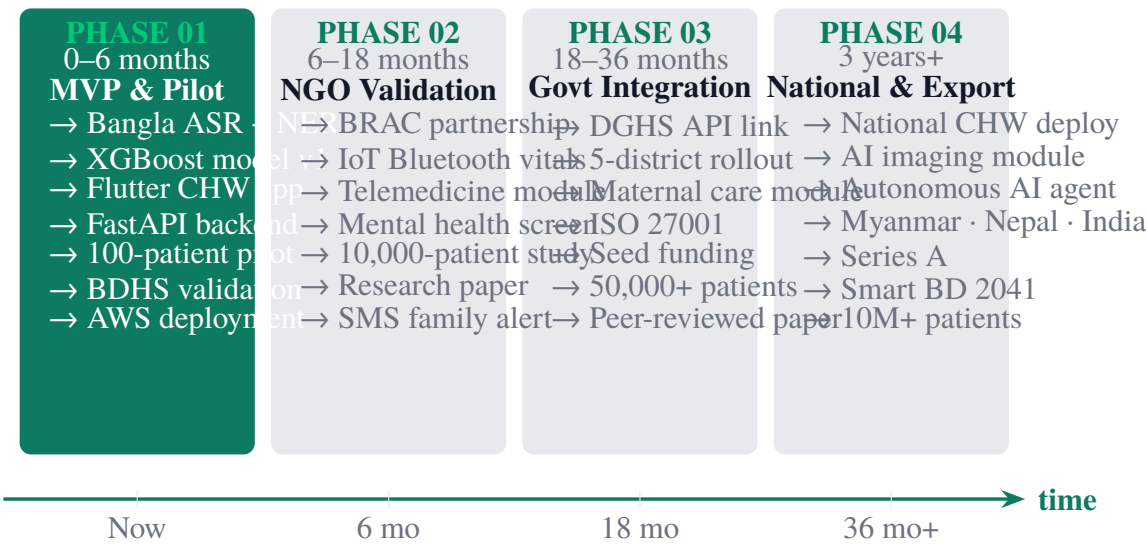


Figure 4. Four-phase development and deployment roadmap

11.2 Export Market Opportunity

The platform is language-agnostic. Expanding to a new country requires only retraining the ASR and NLP layers — the risk model, triage logic, and dashboard remain unchanged.

Table 9. Priority Export Markets — Identical Problem Profile

Market	Doctor Ratio	Rural Pop.	CHW Network
Myanmar	6.1 per 10,000	68% rural	Community Health Workers
Nepal	7.3 per 10,000	80% rural	Female Community Health Volunteers
India (Bihar)	6.5 per 10,000	89% rural	ASHA workers (1M+)
Cambodia	7.2 per 10,000	76% rural	Village Health Support Group

11.3 Long-Term Vision

By 2030, *ShasthoBondhu AI* aims to be the standard AI screening infrastructure for Community Health Workers across South and Southeast Asia — contributing to NCD risk detection in over 10 million people who would otherwise have zero preventive healthcare contact, and positioning Bangladesh as a contributor, not a recipient, of global health AI research.

12. Conclusion

ShasthoBondhu AI is not a symptom checker or chatbot. It is a fundamentally different intervention: designing for the CHW as the primary user, adding Bangla voice input, offline-first AI inference, Bangladesh-specific clinical calibration, and generative personalised guidance closes every gap left by existing solutions.

The working prototype demonstrates that the full pipeline — Bangla ASR, clinical NLP, on-device XGBoost inference, automated triage, Gemini guidance — is functional, validated, and performant on hardware representative of the actual deployment environment.

Projected Impact at Scale

If deployed to 15% of the NCD-risk rural population by Year 3 (6 million patients), and if the 30–45% late-stage NCD reduction from comparable LMIC AI deployments holds:

- Approximately **1.8–2.7 million crisis events prevented** over three years
- **144 crore BDT per year** in sustainable ecosystem revenue
- **Bangladesh positioned as a model** for CHW-AI integration across the Global South

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